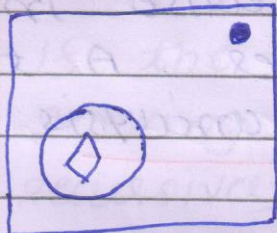
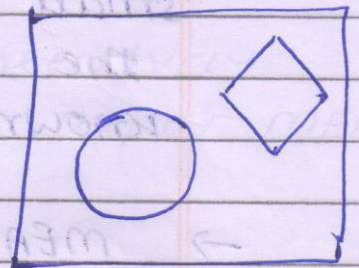
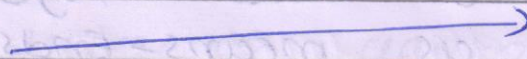


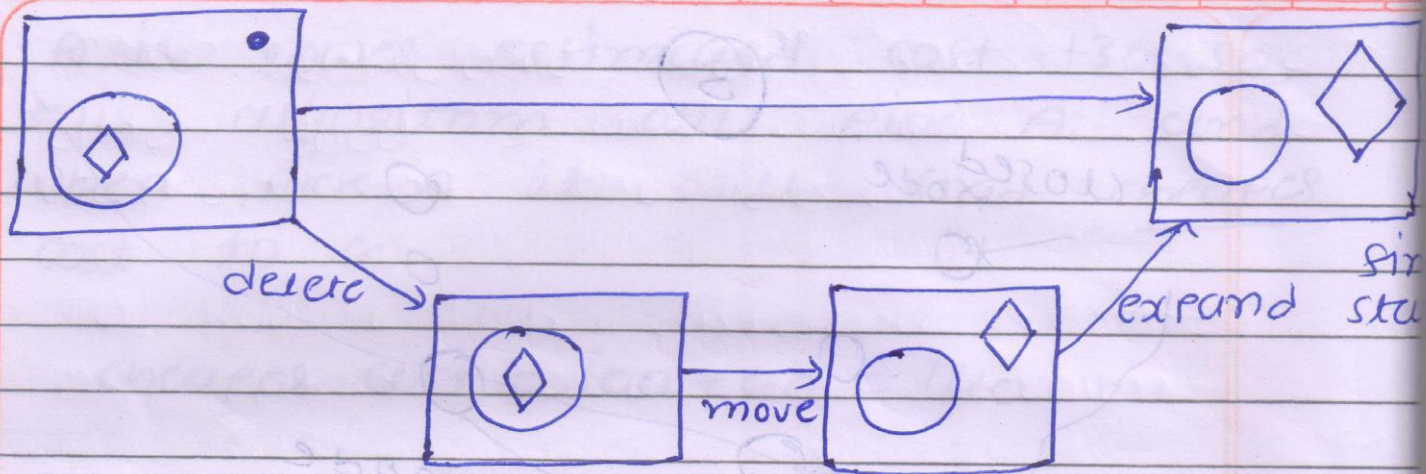
★ Means - Ends Analysis (Problem solving technique)
[MEA]



Initial state



Final state



- Backward + Forward search strategy is used
- such a mixed strategy would make it possible to solve the major parts of the problem first and then go back and solve the small problems that arise in gluing the big pieces together. A technique known as means-Ends-analysis
- MEA process centers around the detection of differences betⁿ current state and goal state
- Use operators that can reduce the difference betⁿ initial and goal state
- Perhaps the operator cannot be applied to current state
 e.g. In above example we cannot use move, and expand first
 → first use delete operator then we can use other operators move then expand

Teacher's Signature.....

→ so, we setup a subproblem of getting to a state in which it can be applied. The kind of Backward chaining in which operators are selected and then subgoals are set up to establish the preconditions of operators is called operation subgoaling.

→ But may be operator does not produce exactly the goal state we want, Then we have a second subproblem of getting from the state it does produce to the goal.

→ if the operator is not effectively reducing difference, again apply MEA recursively.

Ex: - Consider household robot domain

- available operators as shown in fig

- suppose that robot in this domain were given a problem of moving desk with two things on it from one room to another

- objects on top must also be removed

— main difference betⁿ start state and goal state would be the location of the desk

→ To reduce this difference either PUSH or CARRY could be chosen

→ If carry is chosen, its precondition must be met

— location of robot

— size of desk

→ location of robot can handle by walk, but there are no operators that can change size of object

→ so path leads to dead-end

→ we attempt to apply PUSH
four condⁿ

— robot must be at desk

— desk must be clear

— desk is large

— robot's arm empty

Assumption: 2 small objects are resting on desk

Step 1

start

goal

to move
robot to
place)

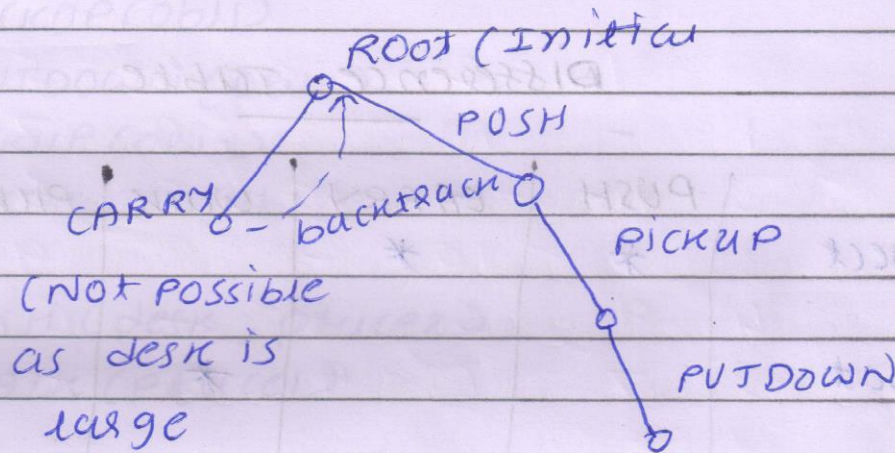
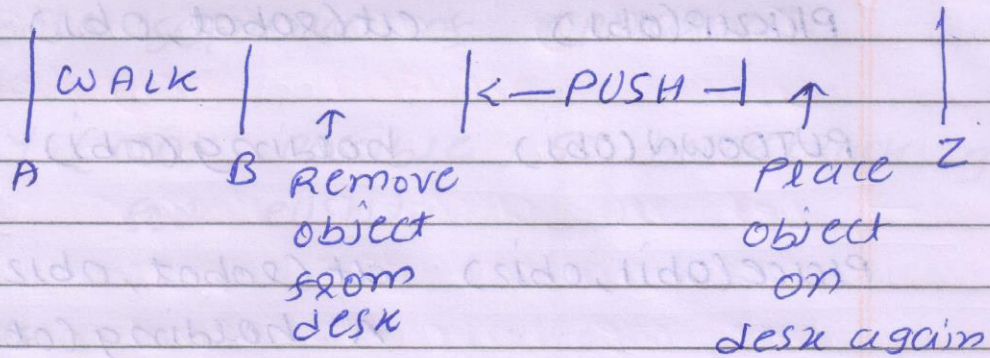
| WALK |
A (place) B

← difference
reduced →

Z

Step 2

PUSH identified
as carry
cannot be
applied



~~Operator~~

Robot's operators

Operator	Preconditions	Results
$PUSH(obj, loc)$	$at(robot, obj) \wedge$ $large(obj) \wedge$ $clear(obj) \wedge$ $armempty$	$at(obj, loc) \wedge$ $at(robot, loc)$
$CARRY(obj, loc)$	$at(robot, obj) \wedge$ $small(obj)$	$at(obj, loc) \wedge$ $at(robot, loc)$
$WALK(loc)$	none	$at(robot, loc)$

Pickup(obj) at(robot, obj) holding(obj)
 PUTDOWN(obj) holding(obj) $\neg$ holding(obj)
 Place(obj1, obj2) at(robot, obj2) on(obj1, obj2)
 $\neg$ holding(obj1)

Difference Table

	PUSH	CARRY	WALK	PICKUP	PUTDOWN	Place
move object	*	*				
move robot			*			
clear object				*		
act object on object						*
act arm empty					*	*
Be holding object				*		

— Difference table that describes when each of operators is appropriate

So, whole sequence of operators

1) WALK

2) CARRY (NOT POSSIBLE, SO BACKTRACK go
FOR PUSH)

~~3) WALK~~

3) PICKUP(OBJ1)

4) PUTDOWN(OBJ1)

5) PICKUP(OBJ2)

6) PUTDOWN(OBJ2)

7) PUSH(desk, place2)

8) WALK(Place1)

9) CARRY(OBJ1, Place2)

10) WALK(Place1)

11) CARRY(OBJ2, Place2)

12) PICKUP(OBJ1)

13) Place(OBJ1, desk)

14) PICKUP(OBJ2)

15) Place(OBJ2, desk)